ingredients 1 2 3
tests 1 2 3

$\frac{\log n}{2(n-1)}$

$\frac{1}{2}\log(4\pi V)$

χ_{zz}, variance per site

$\Delta S = ?$ $V = ?$

iDMRG, the computer

M, exact inverse length

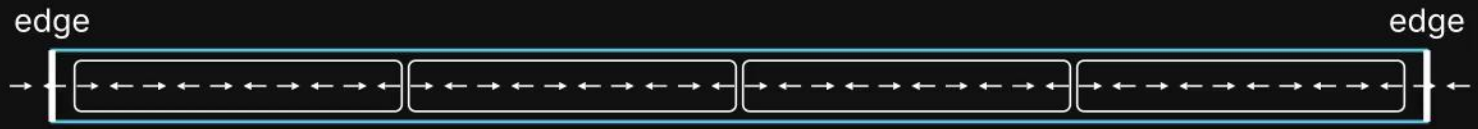
the lightest particles

paper: fig_asym_n, panel (a)
data/figures/fig_asym_n_300dpi.png
Delta_z = 0.3, gamma = 0.2

? PAUSE AND PONDER

For a mixed window, is the Clock's width V equal to the variance of the charge Bell?

Vote: Yes, always | Only when the window is pure | Never for a mixed window





`1/M: paper: jkm_exact.py; Delta_z = 0.3, gamma = 0.2`

chunks longer than 1/M are independent: logs add;
both widths grow by the same amount per chunk;
only the edges differ

BLACK BOX · `transfer matrix ("logs add")` $\log Z_2(\alpha), \quad \log\langle e^{i\alpha Q_A}\rangle$

- For the chain's ground state as the computer stores it (3.12), both of these logs grow linearly with ℓ, and their curvatures at α = 0 give the same width per site: chunks longer than 1/M are independent, and the logs of independent pieces add.
- Both rates come from one object, the leading eigenvalue of a phase-dressed transfer matrix; what is left between V and the Bell's variance is an edge term that does not grow with ℓ.
- Proved in the paper for free fermions and for the computer's class of states (App. A); taken on faith here.

ingredients 1 2 3
tests 1 2 3

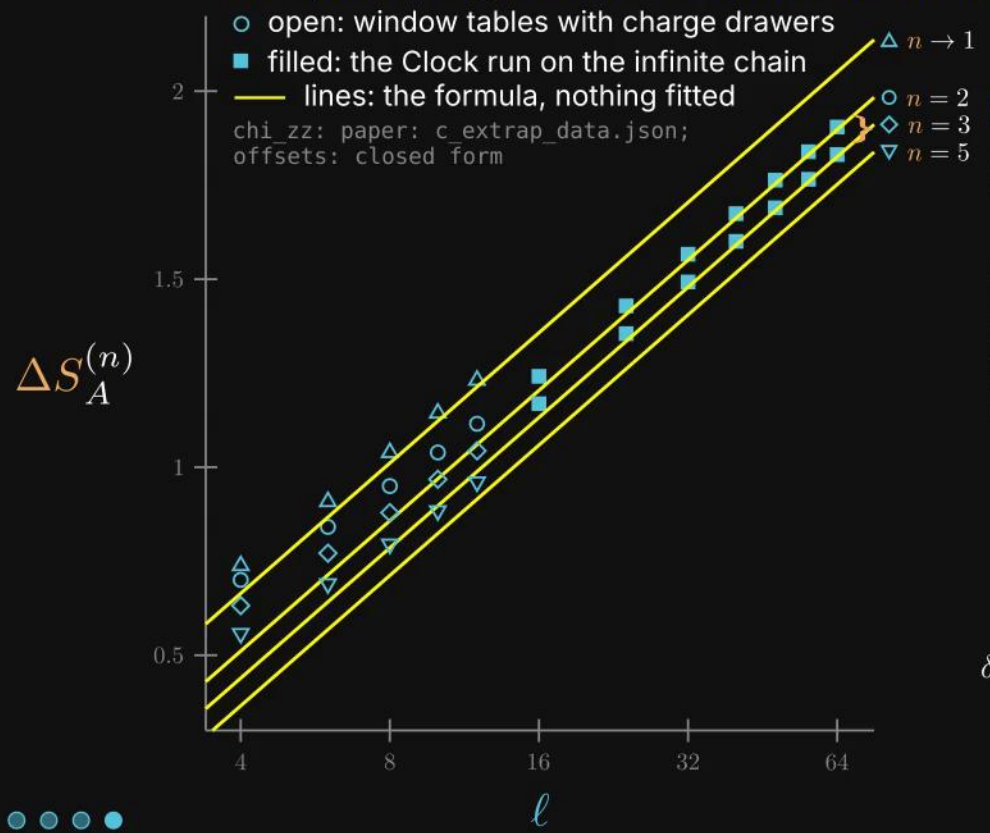
$$\Delta S_A^{(n)}(\ell) = \tfrac{1}{2}\log(\pi \ell \chi_{zz}) + \frac{\log n}{2(n-1)} + \mathcal{O}\left(\frac{1}{M\ell}\right)$$

Blue and yellow: half the log of pi, times the window, times the variance per site.
Gold: a pure number from the Dial; one half at n to 1, 0.3466 at n = 2. `0.3466: closed form`
White: the edge term, a kink's length over the window's length.
Nothing fitted; on faith: “logs add” (4.2) and the walk from integer n to n = 1 (Black-box card).

BLACK BOX · `the n → 1 continuation`

- Card again from 2.11: the gold offset log n / (2(n – 1)) is derived for every integer n ≥ 2, a Gaussian integral over n replicas placed on a circle.
- The von Neumann value is that formula continued to n → 1, where it reads 1/2; it is checked against the n → 1 points of Fig. 2(a) in 4.4, not derived.
- Behind it: all replica widths are equal, V_d = V; proved for the computer's class of states, with an edge term that stays finite as the window grows.

Test 1 passed: slope one half, offsets from the Dial, nothing fitted.

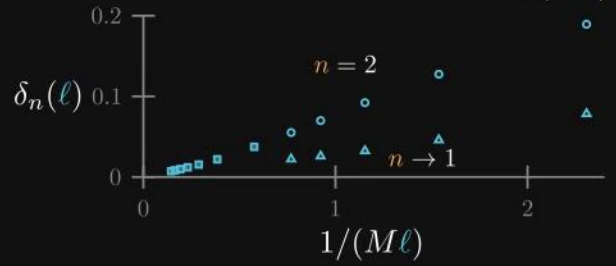


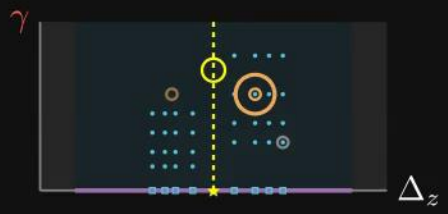
$0.0719 = \frac{1}{2}\log 2 - \frac{1}{4}\log 3$ closed form

measured on the same state:
the same gap, within three percent

paper: rdm_030_020.json n2 n3 n5 vN
(open, ell <= 12); asym_production.json
curve (filled, n = 2, ell = 16-56);
n3_results.json interacting,
interacting_ext (n = 3 rows, ell = 64);
M = 0.10836 from jkm_exact.py;
Delta_z = 0.3, gamma = 0.2

$$\delta_n(\ell) \equiv \Delta S_A^{(n)}(\ell) - \left[\tfrac{1}{2}\log(\pi \ell \chi_{zz}) + \frac{\log n}{2(n-1)}\right] = \mathcal{O}\left(\frac{1}{M\ell}\right)$$





$$\chi_{zz} = 0.1106 \quad \boxed{0.25\ \%} \quad \chi_{zz} = 0.1103$$

$$c_1 = \tfrac{1}{2}\log(2\pi\chi_{zz}) \;\Rightarrow\; \chi_{zz} = \frac{e^{2c_1}}{2\pi}$$

$$c_1 < 0 \text{ because } \chi_{zz} < \frac{1}{2\pi} = 0.159$$

attractive repeat, Δz = −0.3, γ = 0.2: c_1 measured 0.2111, formula 0.2133
free line, Δz = 0, γ = 0.25: exact χzz = γ/(1+γ) = 0.2, asymmetry road 0.1998

At finite γ the free line reads χzz/M = 0.78 here (Δz = 0, γ = 0.25). The star value 1 is the γ → 0 limit; the road to it is scene 4.10.

paper: c_extrap_data.json (direct chi_zz) and asym_production.json c1_inf (Delta_z = 0.3, gamma = 0.2); val2.json chizz, c1_inf, pred (Delta_z = -0.3, gamma = 0.2); n3_results.json anchor (Delta_z = 0, gamma = 0.25); exact XY value: closed form
0.1106: iDMRG, infinite chain
0.1103, 0.25 %, 0.159, 0.78: closed form

χ_{zz}  the budget

$$\chi_{zz} = \frac{2}{L} \sum_{k \neq 0} \frac{|\langle k|\dot{Q}|0\rangle|^2}{E_k^2}$$

$\dot{Q} = i[H, Q] \neq 0$
the charge leaks

Every bar is positive: keep a few, and you hold a lower bound.

BLACK BOX · the sum rule

- Every excited state of the chain gets one bar; its height is the leak rate into that state, squared, divided by the state's energy squared; every bar is positive.
- The step from "variance" to "leak over energy" is one line: for an eigenstate, the matrix element of $\dot{Q}$ is iE_k times that of Q (Ledger).
- Which states are the lightest with a bar is decided by the dictionary of 3.10b; we take that list (two walls flying apart, and the pink rung) on faith; heavier states add more positive bars.





$$\mathfrak{c}(\beta^2) \equiv \chi_{zz}/M$$

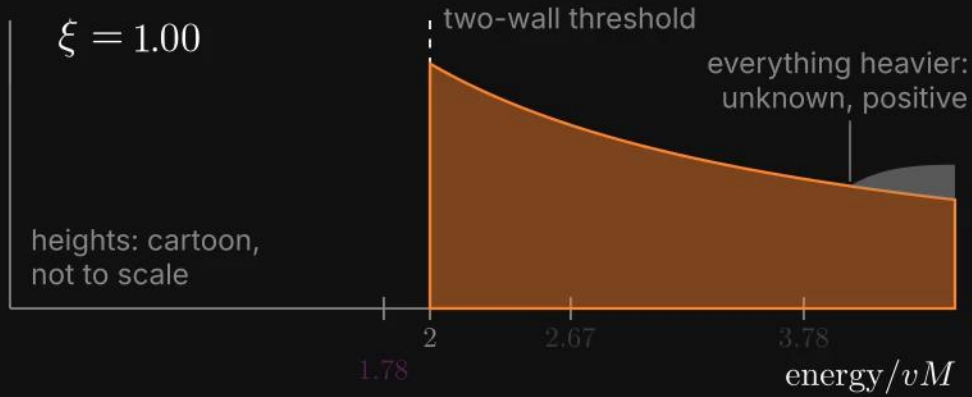
named in 4.8b

$$\geq \mathfrak{c}_{2k} + \mathfrak{c}_{B_1}\Theta(1-\xi)$$

A lower bound, not a prediction.

```
closed form: 1.78 = 2 sin(0.35π), 2, 3.78 = 2 + 1.78
hump: data/hump.npz h, peak at E = 2.67 vM (ξ = 0.70)
ξ = 0.70 is Δz = −0.2737 (closed form, γ → 0)
```

ingredients 1 2 3
tests 1 2 3





$$\mathfrak{c}(\beta^2) \equiv \chi_{zz}/M$$

named in 4.8b

$$\geq \mathfrak{c}_{2k} + \mathfrak{c}_{B_1}\Theta(1-\xi)$$

A lower bound, not a prediction.

closed form: scripts/gen_hump.py (data/hump.npz) = paper: closed_form_results.json master_table; square: fig1_points.json orig[1]; ξ = 0.70 is Δz = −0.2737, γ → 0

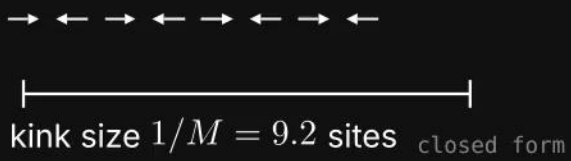
BLACK BOX · a function Lukyanov found

- The orange hump's shape is a function Lukyanov found in 1997 for the two-wall states of the continuum dictionary of 3.10b; its overall size is fixed by one more identity and one constant matched at the free point, not fitted.
- That identity is Hellmann-Feynman: the derivative of the ground-state energy with respect to the breaking coupling trades an unknown vacuum average for the inverse length M and the coupling, and reproduces a known exact energy.
- The pink bar's height is Lukyanov's one-breather value; its weight vanishes as ξ → 1 and its slope cancels the hump's threshold kink, a check the paper performs.

$$M = \operatorname{atanh} k_1'$$

k_1': an elliptic modulus (App. C, Ledger)





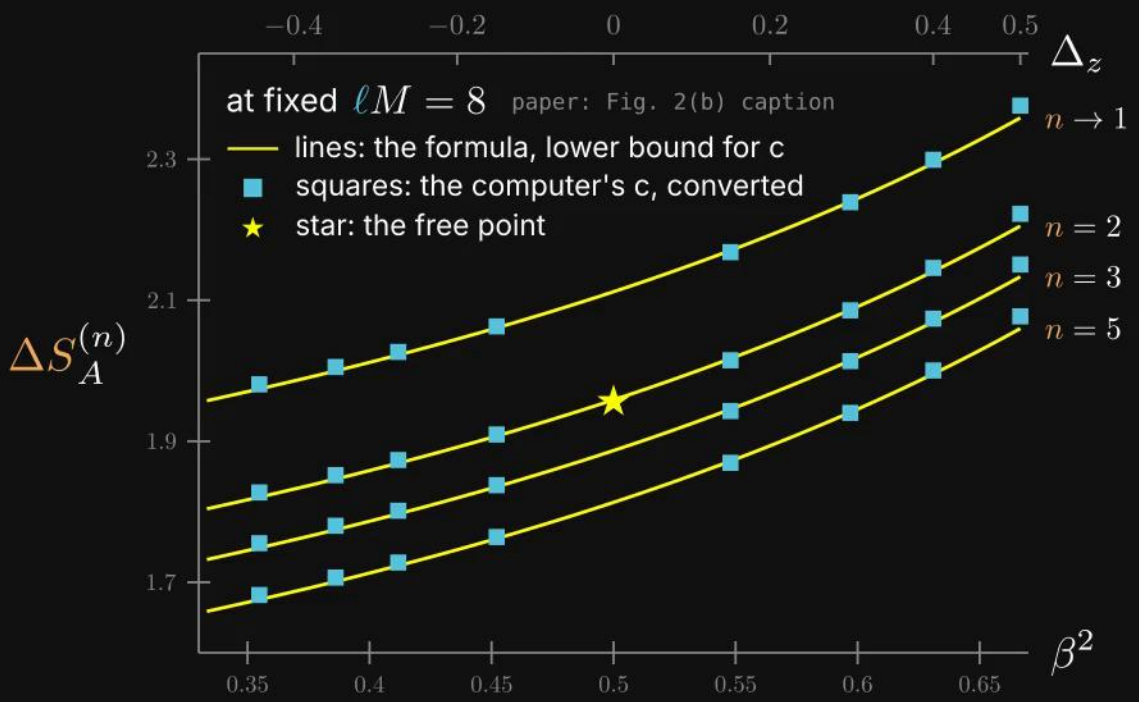
closed form: jkm_exact.py
M_exact(g, Dz) (data/masses.npz);
exponents paper:
C_amplitude_results.json p;
Δz = 0, 0.3, 0.5, γ → 0

BLACK BOX · Baxter / JKM elliptic functions

- Card again from 3.7: the machine is the exact solution of the eight-vertex model (Baxter 1972), read for the XYZ chain by Johnson, Krinsky and McCoy (1973); k_1' is an elliptic modulus built from Δz and γ (App. C).
- On the XY line every elliptic function reduces to one line and M = atanh γ; at small γ the output follows $M \propto \gamma^{(\pi/2\theta_0)}$ with a known prefactor (4.8b).
- We use only the output, jkm_exact.py at 20 digits; never a fitted mass.

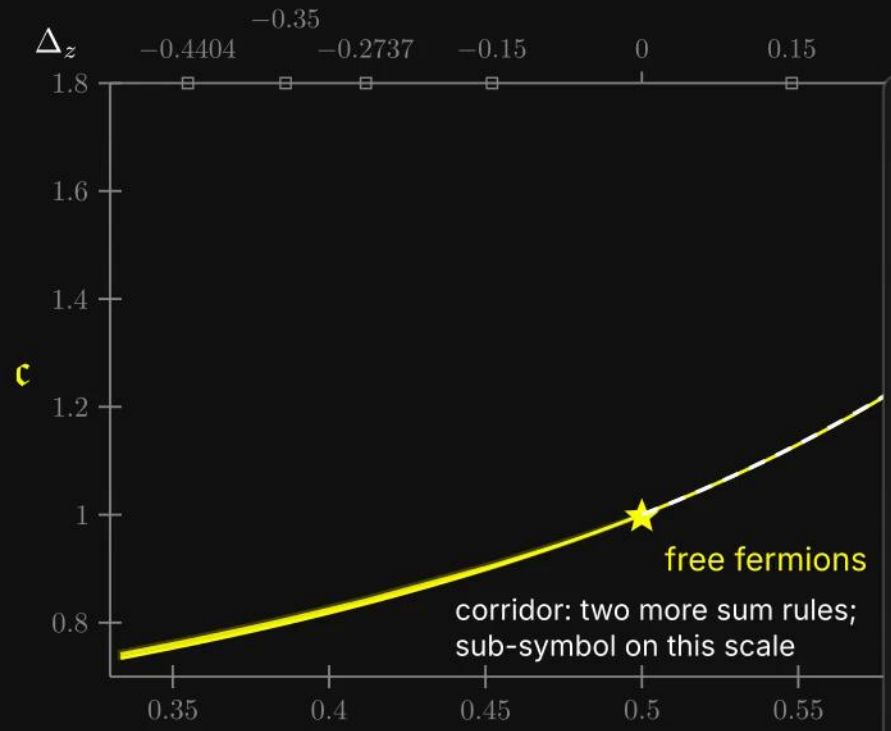
$$\Delta S_A^{(n)} = \tfrac{1}{2}\log(\pi\ell) + \tfrac{1}{2}\log\left[\mathfrak{c}(\beta^2)\left(\frac{\gamma\, 4^{1-2\beta^2}}{\sin^2\theta_0}\right)^{\pi/2\theta_0}\right] + \frac{\log n}{2(n-1)}$$

ingredients 1 2 3
tests 1 2 3



Eq. (final) holds for γ → 0 at fixed ℓM ≫ 1. Finite-γ corrections: a regular power series in γ, and a lattice-only fractional power $\gamma^{(4K-2)\pi/2\theta_0}$ with $K = \frac{1}{2\beta^2}$. Named here, used in the error bars (4.10), never shown.

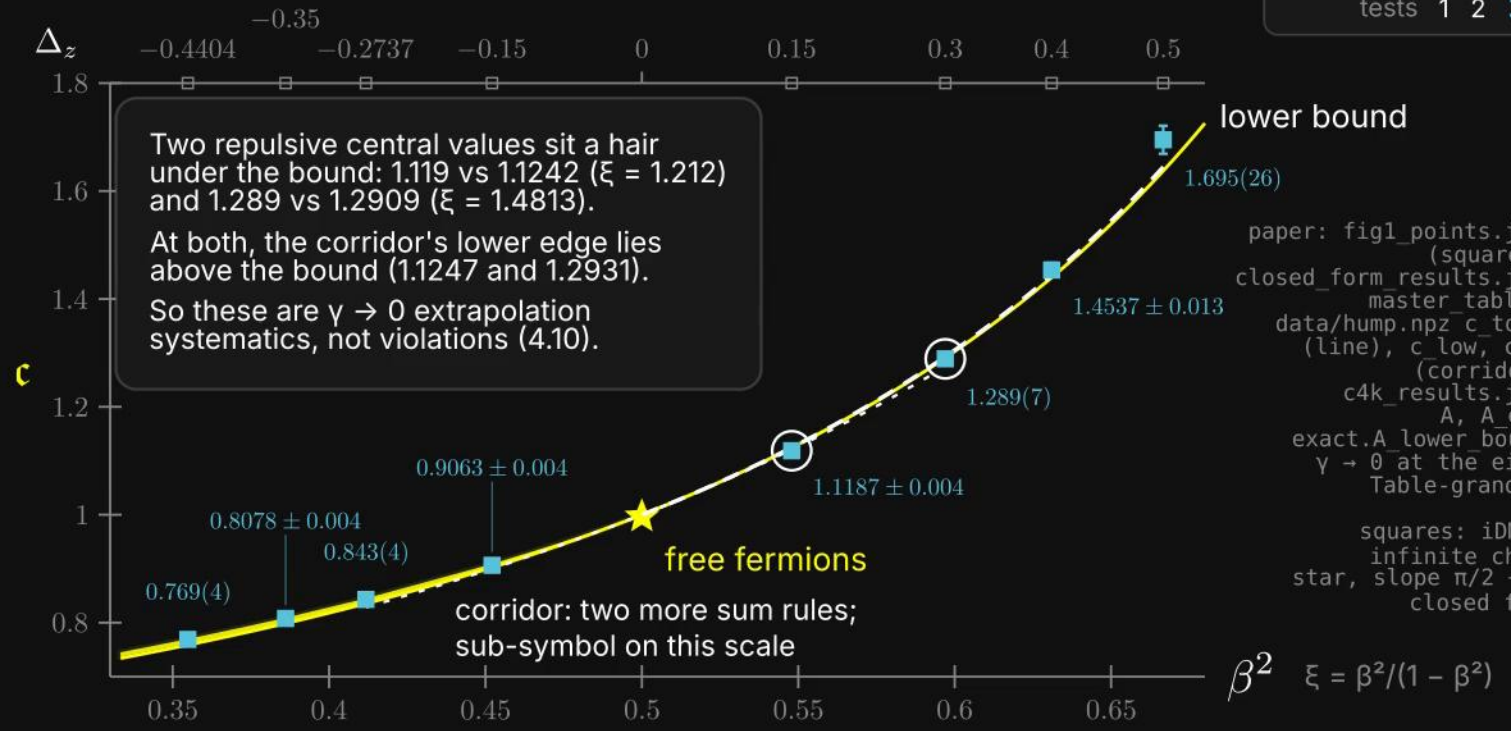
paper: closed_form_results.json master_table (lines, c2k + cB1 on the ξ grid); fig1_points.json (squares); C_amplitude_results.json p; ℓM = 8, γ → 0



BLACK BOX · the moment corridor

- Two more sum rules weigh the same bars with $1/E_n$ and with E_n; both totals are known in closed form (a shift of the field for the first; an f-sum rule for the second).
- Subtract the bar and the hump from both, and what is left are two moments of the unknown tail; the lightest omitted threshold gives a lower edge on both sides, and Cauchy-Schwarz an upper edge, which survives only for $\xi < 1$, where the E_n total is finite.
- The closed forms and the tail moments live in App. B and on the Ledger; the band on screen is their output.

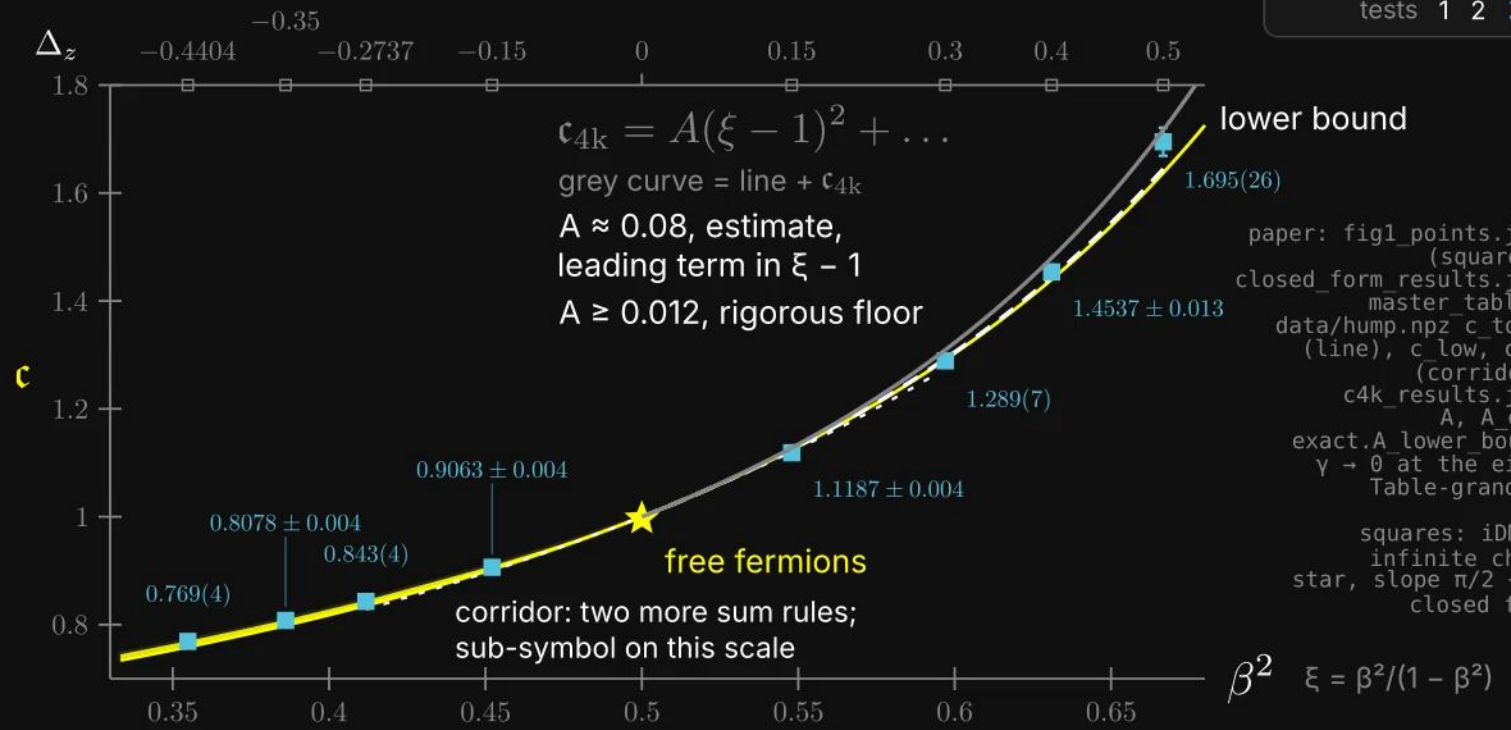
ingredients 1 2 3
tests 1 2 3



paper: fig1_points.json (squares); closed_form_results.json master_table = data/hump.npz c_total (line), c_low, c_up (corridor); c4k_results.json A, A_err; exact.A_lower_bound; γ → 0 at the eight Table-grand Δz

squares: iDMRG, infinite chain
star, slope π/2 − 1: closed form

ingredients 1 2 3
tests 1 2 3



paper: fig1_points.json (squares); closed_form_results.json master_table = data/hump.npz c_total (line), c_low, c_up (corridor); c4k_results.json A, A_err; exact.A_lower_bound; γ → 0 at the eight Table-grand Δz

squares: iDMRG, infinite chain
star, slope π/2 − 1: closed form

Test 3: a lower bound honoured within the uncertainties at eight couplings; the corridor holds at all four attractive ones.

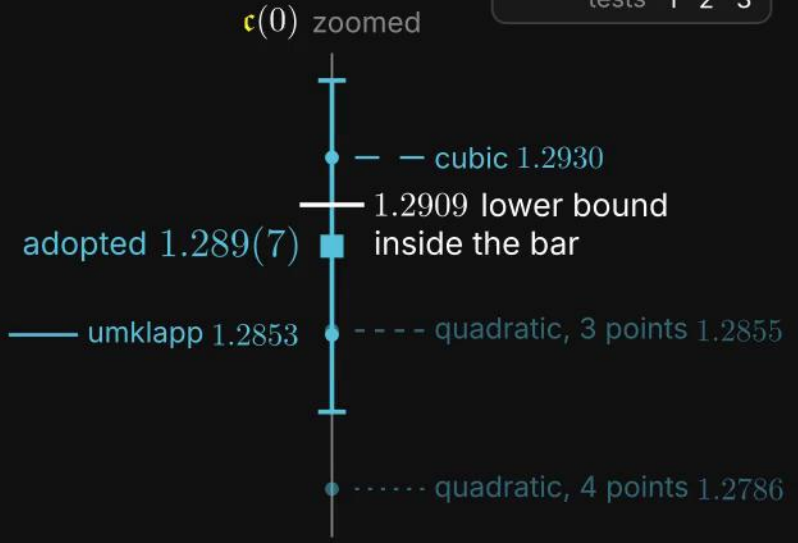
$$\mathfrak{c}(\gamma) = \mathfrak{c}_0(1 + a\gamma + b\gamma^s)$$

BLACK BOX · the γ → 0 extrapolation forms

- Four roads: a quadratic through the first three points, a quadratic through all four, a cubic through all four, and the umklapp form (formula above) with the exponent s fixed by Δz (no free exponent).
- Adopted value: the mean of the cubic and umklapp roads; error bar: half the spread of the four.
- Why a fractional power at all: lattice operators absent from the continuum; the exponent is $(4K-2)\pi/2\theta_o$ with $K = 1/(2\beta^2)$ (Ledger).

iDMRG, infinite chain; paper: jkm_adjudication.json c_vals, quad3, quad4, cubic, phys (Δz = 0.3); chi_conv.json (Δz = 0.5, γ = 0.1); bound closed_form_results.json at ξ = 1.4813

ingredients 1 2 3
tests 1 2 3

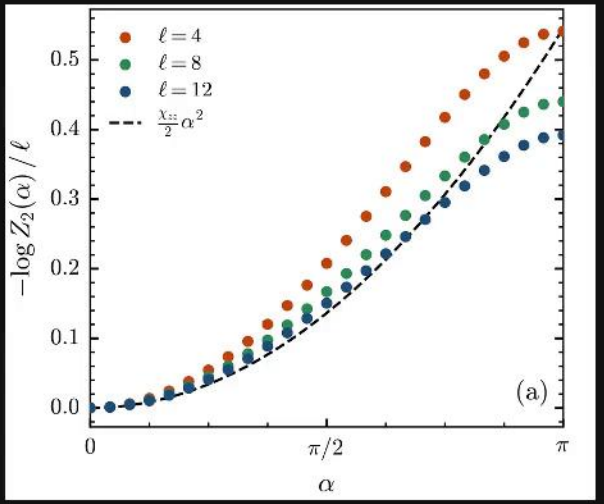


D	χ_{zz}
64	0.033726
96	0.033723
128	0.033723

Δz = 0.5, γ = 0.1, the hardest point; paper: chi_conv.json; iDMRG, infinite chain

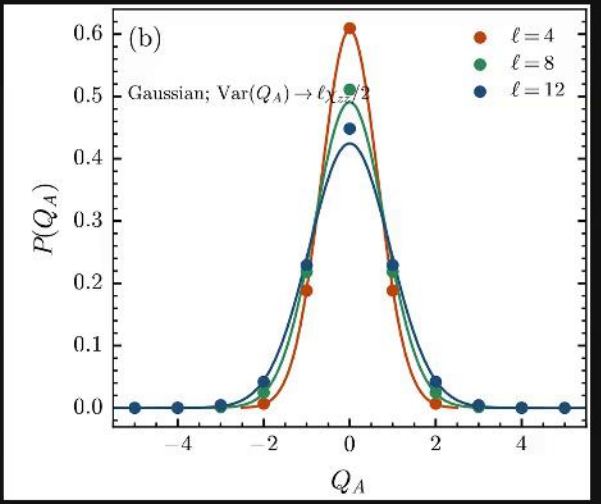
the computer's own error is far smaller than the road's

ingredients 1 2 3
tests 1 2 3



the coins





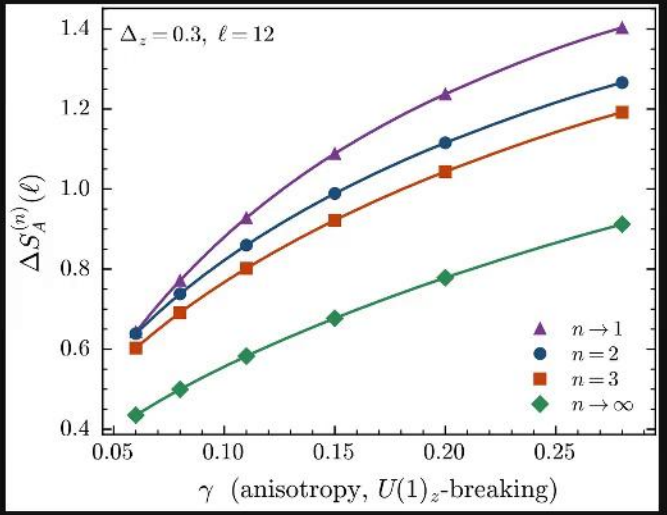
$$-\frac{\log Z_2(\alpha)}{\ell} \to \frac{\chi_{zz}}{2}\alpha^2$$

the Gaussian saddle of 2.10, measured

solid: bells of the measured variance, which tends to $\ell\chi_{zz}/2$

paper: fig_fcs (PNG)
data/figures/fig_fcs_300dpi.png
Δz = 0.3, γ = 0.2

ingredients 1 2 3
tests 1 2 3



top to bottom: $n \to 1,\ 2,\ 3,\ \infty$

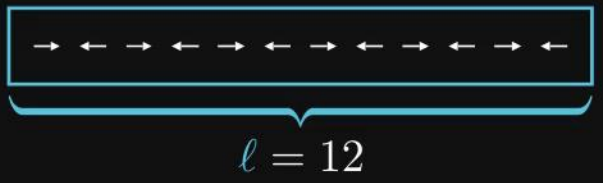
paper: fig_asym_gamma (PNG, data/figures/fig_asym_gamma_300dpi.png), Delta_z = 0.3, ell = 12, gamma sweep

ruler: 1/M, closed form; M from jkm_exact.py M_exact(g, Dz=0.3)

$1/M = 21.7$ sites

$\gamma = 0.10$





$\ell = 12$

Fixed ℓ: the meter reads 0 at γ = 0 and rises with γ; the formula holds where 1/M < ℓ.
two limits, ℓ → ∞ and γ → 0, do not commute; this plot is the lattice echo of that

TAKE-HOME

$$\Delta S_A^{(n)} = \tfrac{1}{2}\log(\pi \ell \chi_{zz}) + \frac{\log n}{2(n-1)}$$

- The meter of Part 2 on the chain of Part 3 reads ½ log(πℓχzz) + log n / 2(n−1) plus an edge term of order 1/(Mℓ): half the log of the variance, nothing fitted, three tests passed.
- The variance per site is a budget spent over excited states; the two lightest particles, two walls and the pink rung, give a lower bound on χzz/M with no free parameter, and a corridor pins the rest on the attractive side.
- With Baxter's exact inverse length the formula closes in lattice variables (Eq. final): γ and Δz in, the asymmetry out, valid for γ → 0 and ℓM ≫ 1; the error bars are half the spread of four roads to γ → 0.

LEDGER

- ☐ Eq. (V) was shown once; V = Var(Q in the window) × [1 + O(ℓ^{-1})] was taken on faith (“logs add”), proved in the paper for free fermions and injective matrix-product states only (App. A) (4.2).
- ☐ Which column of V_var_exhibit.json is V and which the window's variance is not established; only the common slope and the constant gap were used (4.2).
- ☐ Still on faith from Part 2: the 2023 definition and the quantum Mpemba effect, named only (2.1); $\Delta S^{(n)} \geq 0$, asserted, its one-line majorization proof on Part 2's Ledger (2.5).
- ☐ Still on faith from Part 3: Jordan-Wigner, “XY line = free fermions” (3.1b); gap ⇒ exponential decay, Hastings-Koma 2006 (3.7); the continuum dictionary of 3.10b and its bound-pair law; everything inside iDMRG, the computer behind every filled point and every square (3.12).
- ☐ The Dial's offsets come from the replica-loop Gaussian integral (2.11) and assume equal replica widths, V_d = V, proved for the computer's class of states; the n → 1 value is an analytic continuation checked against data (2.11, 4.3, 4.4).

- ☐ The sum rule's one-line step from the variance to "leak rate squared over energy squared" was stated, not derived (4.6); which excited states carry a bar (two walls in the odd channel, the pink rung; not B_1B_1 or B_2) comes from the continuum dictionary of 3.10b; the hump's shape is "a function Lukyanov found" and was never written out; the Hellmann-Feynman step eliminates a vacuum average (4.6, 4.7).
- ☐ The line of Fig. 1 is a lower bound; the corridor needs two more sum rules whose closed forms were not shown; it is two-sided only for $\xi < 1$ (4.7, 4.9).
- ☐ $A \simeq 0.08$ is a Monte-Carlo estimate with about 20 % systematics; $A \geq 0.012$ is the rigorous floor; the letter A here is the four-kink coefficient, not the window of 1.2 (4.9).
- ☐ M comes from elliptic functions (Baxter/JKM) we never opened; $M \propto \gamma^{(\pi/2\theta_o)}$ is its $\gamma \to 0$ limit and the finite-γ slopes differ; M is an inverse length, a kink's energy is vM (4.8).
- ☐ Eq. (final) holds for $\gamma \to 0$ and $\ell M \gg 1$; finite-γ corrections (a power series and a lattice-only fractional power) enter only through the extrapolation (4.8b, 4.10).

```
A ~ 0.08: paper: c4k_results.json A
(an estimate, ~20 % systematics, expansion about ξ = 1)
A ≥ 0.012: paper: closed_form_results.json exact.A_lower_bound
```

- ☐ Error bars are half the spread of four fit forms with the mean of cubic and umklapp as centre; the two marginal repulsive points of Fig. 1 are extrapolation systematics and stay out of the finale (4.9, 4.10).
- ☐ On the free line at $\gamma = 0.25$ the ratio χ_{zz}/M is 0.78, not 1; the star is the $\gamma \to 0$ limit (4.5).
- ☐ Fig. 3 and Fig. 5 are the published PNGs; their arrays are not in the repository (4.11, 4.12).
- ☐ The anchor coupling $\Delta z = 0$, $\gamma = 0.25$ behind the free-line files is inferred, not stored (4.5).
- ☐ Four helper symbols beyond the roadmap's four Part 4 symbols (V, owned by 2.9 and extended to mixed windows in 4.2; c, c_1, β^2): $\delta_n(\ell)$ (4.4), $\dot{Q}$ (4.6), $\theta_o = \arccos \Delta z$ (4.8, a Ledger symbol of 3.10b) and the four-kink A (4.9); each sits in its scene's roadmap equation and in the handout glossary.



0.78 = (0.25/1.25)/atanh(0.25): closed form
(χzz = γ/(1+γ), M = atanh γ; Δz = 0, γ = 0.25)

Beyond the Neel-x phase: $\Delta z > 1$, where the breaking is staggered and the same marginality returns.

A field coupled to the broken charge.

After a quench: Eq. (final) is the equilibrium baseline for Mpemba-type behaviour.

The tail: resum the four-kink term; a convergent repulsive-side f-sum rule would close the corridor.

What does the meter read beyond the Neel-x phase, in a field, or after a quench? That is the research frontier.